Answer the Following Questions:

Question (1) Multiple Choice

Identify the letter of the choice that best completes the statement or answers the question. ***(8 Marks)***

____ 1. Last host of 10.0.0.0/8?
a. 10.0.0.0 b. 10.255.255.254 c. 10.255.255.255 d. 10.0.0.1

____ 2. An organization uses IP 10.5.6.7. How large is its default network?
a. 256 b. 65536 c. 16777216 d. 16777214

____ 3. First host of 10.0.0.0/8?
a. 10.255.255.254 b. 10.255.255.255 c. 10.0.0.0 d. 10.0.0.1

____ 4. The network layer is responsible for delivery of individual............. from source to destination
a. payload b. packets c. bits d. frames

____ 5. What is the speed of Fast Ethernet?
a. 100 Mbps b. 10 Gbps c. Gbps d. 10 Mbps

____ 6. Which of the following is an example of a logical address?
a. 00:1A:2B:3C:4D:5E b. COM1 c. 192.168.1.1 d. 80:00:20:0A:B1:2D

____ 7. Determine the starting address of the block containing 10.5.6.7
a. 10.5.6.0 b. 10.255.255.255 c. 10.5.0.0 d. 10.0.0.0

____ 8. Broadcast of 192.168.5.64/26?
a. 192.168.5.255 b. 192.168.5.64 c. 192.168.5.127 d. 192.168.5.63

____ 9. The Address Resolution Protocol (ARP) is primarily used to resolve:
a. Port addresses to IP addresses b. IP addresses to MAC addresses c. MAC addresses to domain names d. Logical addresses to hostnames

____ 10. What is the ending address for the block of 10.5.6.7?
a. 10.0.0.0 b. 10.5.255.255 c. 10.5.6.255 d. 0.255.255.255

____ 11. Which address type uniquely identifies a network interface on a LAN?
a. Port Address b. Logical Address c. Specific Address d. Domain Address

____ 12. A device has IP 172.16.5.4. How many IPs are in its network?
a. 65536 b. 256 c. 65534 d. 16777216

____ 13. The .... alerts the receiving system to the coming frame and enables it to synchronize its input timing
a. CRC b. The logical Address c. SFD d. Preamble

____ 14. The .... warns the station or stations that this is the last chance for synchronization.
a. Source Address b. DNS c. SFD d. Preamble

____ 15. Which of the following is NOT a function of Ethernet?
a. Routing between networks b. Addressing c. Error Detection d. Data Transmission

____ 16. .... was used to connect two segments of a LAN to overcome the length restriction of the coaxial cables.
a. Router b. Gateway c. Repeater d. Firewall

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	B	C	D	D	A	C	D	C	B	D	B	A	D	C	A	C

Question 2 *(10 Marks)*

a) *For a fast ethernet LAN, How long does it take to create the smallest frame? Show your calculation.* *(2 marks)*

The minimum Ethernet frame size is:

- 64 bytes = 512 bits For Fast Ethernet: Data rate = 100 Mbps
- Time to transmit 512 bits:

$$\text{Transmission Time} = \frac{\mathbf{512}}{\mathbf{100 \times 10^6}}$$
$$= \mathbf{5.12 \times 10^{-6}} \text{ seconds}$$
$$= \mathbf{5.12\ \mu s}$$

Since collision detection requires the signal to travel to the farthest node and back (round trip), the maximum one-way propagation time is:

$$\frac{\mathbf{5.12\ \mu s}}{\mathbf{2}} = \mathbf{2.56\ \mu s}$$

Final Answer

- Maximum round-trip propagation time: 5.12 µs
- Maximum one-way propagation time: 2.56 µs

These limits ensure collisions can still be detected correctly in a CSMA/CD Fast Ethernet network.

b) *Given a scenario where a user is unable to access a website, analyze how each layer of the OSI model could be used to troubleshoot the issue. Which layer would you check first and why?*

(4 Marks)

We are starting from layer 7 (Application Layer)

o Does the protocol (HTTP/HTTPS) is already working? (Application Layer)?

o Is there any response from the server (Transport Layer)?

o Does the IP address was analyzed in a real way (Network Layer)?

o Does the network interface card (NIC) is already working (Data Link & Physical Layers)?

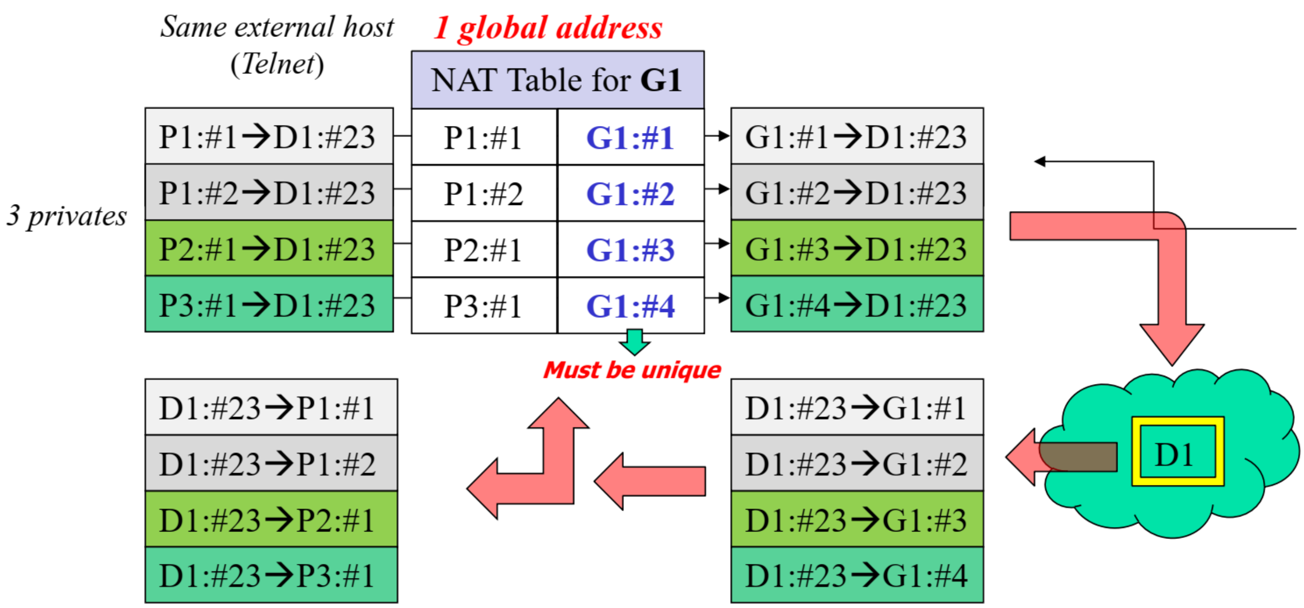
o Clarify the importance of each layer in diagnosing the problem if it is (software or hardware or config).

o According to the problem type, we have to select the first layer to start with (for example if there is a DNS error, we start from Network Layer).

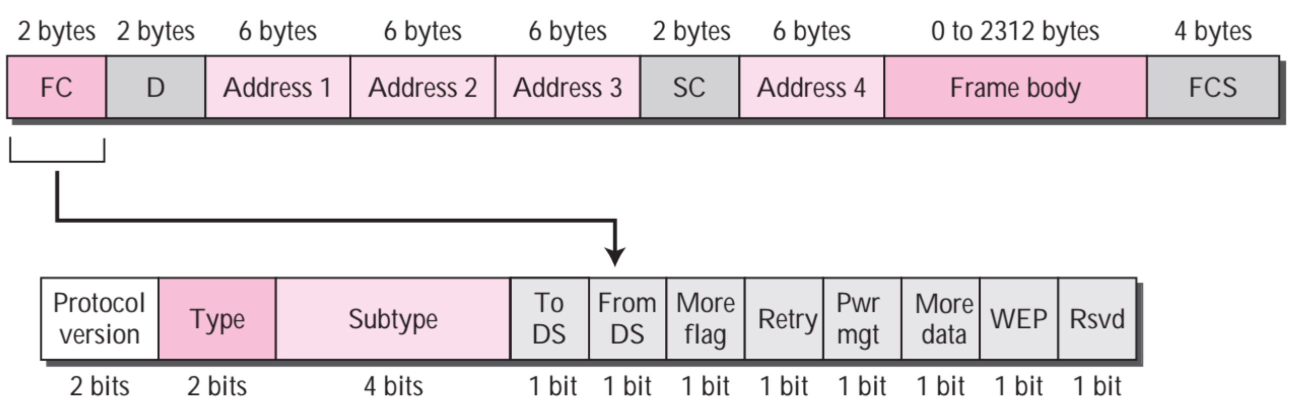
Question (3) **(12 Marks)**

(a) The figure below illustrates a Network Address Translation (NAT) process where multiple private hosts (p1,p2, and P3) communicate with the same external server using a single global IP address, assume the destination host (D2) has a port number #25, the port numbers for Global Address are #1,#2,#3 and #4.

- *Construct the NAT translation table for the connections shown in the figure.*
- *Identify the Inside Local and Inside Global addresses for each private host.* *(4 Marks)*



(b) Explain with aid of figure the structure of a wireless LAN frame format, state also the frame categories defined by IEEE 802.11. *(4 Marks)*



- **D.** In all frame types except one, this field defines the duration of the transmission that is used to set the value of NAV. In one control frame, this field defines the ID of the frame.
- **Addresses.** There are four address fields, each 6 bytes long. The meaning of each address field depends on the value of the *To DS* and *From DS* subfields and will be discussed later.
- **Sequence control.** This field defines the sequence number of the frame to be used in flow control.
- **Frame body.** This field, which can be between 0 and 2312 bytes, contains information based on the type and the subtype defined in the FC field.
- **FCS.** The FCS field is 4 bytes long and contains a CRC-32 error detection sequence.

Frame Types

A wireless LAN defined by IEEE 802.11 has three categories of frames: management frames, control frames, and data frames.

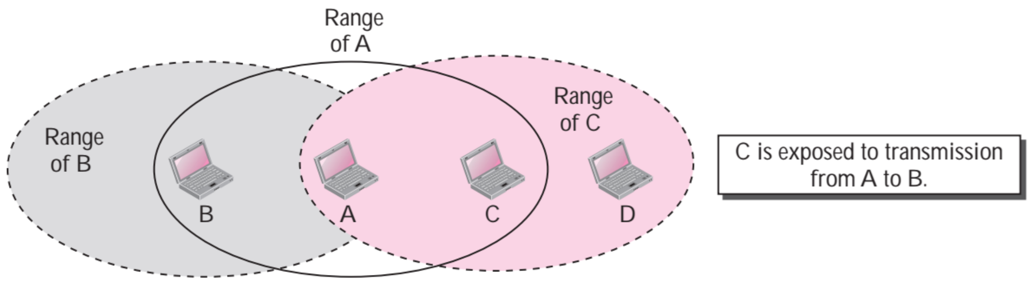
Management Frames Management frames are used for the initial communication between stations and access points.

Control Frames Control frames are used for accessing the channel and acknowledging frames.

Data Frames Data frames are used for carrying data and control information.

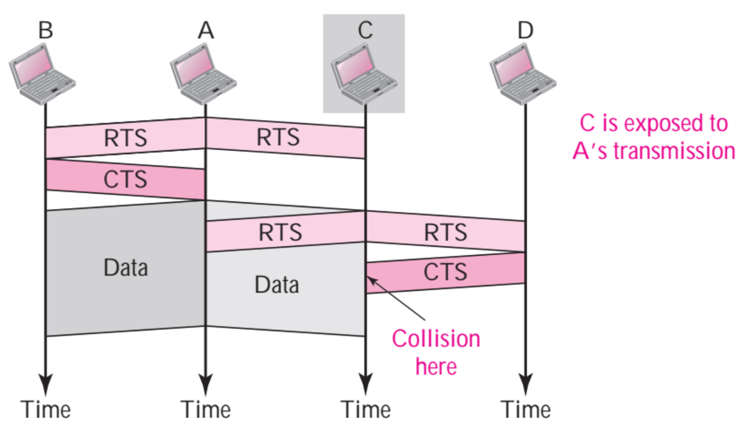
(c) Explain with the aid of figures the exposed station problem where a station refrains from using a channel when it is, in fact, , available. show by graph what happens if handshaking messages RTS and CTS are used in this situation? *(4 Marks)*

Exposed Station Problem Now consider a situation that is the inverse of the previous one: the exposed station problem. In this problem a station refrains from using a channel when it is, in fact, available. In Figure 3.21, station A is transmitting to station B. Station C has some data to send to station D, which can be sent without interfering with the transmission from A to B. However, station C is exposed to transmission from A; it hears what A is sending and thus refrains from sending. In other words, C is too conservative and wastes the capacity of the channel.



The handshaking messages RTS and CTS cannot help in this case, despite what we might think.

Station C hears the RTS from A, but does not hear the CTS from B. Station C, after hearing the RTS from A, can wait for a time so that the CTS from B reaches A; it then sends an RTS to D to show that it needs to communicate with D. Both stations B and A may hear this RTS, but station A is in the sending state, not the receiving state. Station B, however, responds with a CTS. The problem is here. If station A has started sending its data, station C cannot hear the CTS from station D because of the collision; it cannot send its data to D. It remains exposed until A finishes sending its data.



Use of handshaking in exposed station problem

Question (4) (10 Marks)

a) *An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to Four groups of customers as follows:*

   a. The first group has 64 customers: each need 256 addresses.
   b. The second group has 128 customers: each need 128 addresses.
   c. The third group has 128 customers: each need 64 addresses.
   d. The Fourth group has 256 customers: each need 64.

   - *Design the subblocks and find out how many addresses are still available after these allocations.*
   - *Draw a figure showing the four subnetworks. in each network, show the first host address, the last address, and the subnetwork address.*

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then 32 − 8 = 24. The addresses are

1st customer 190.100.0.1 → 190.100.0.254

Last customer 190.100.63.0 → 190.100.63.254
Network Id 190.100.0.0/24
Total =64X256 = 16384

Group 2

For this group, each customer needs 128 addresses. This means that 7 (log2 128) bits are needed to define each host. The prefix length is then 32 − 7 = 25. The addresses are

1st customer 190.100.64.1 → 190.100.64.127

Last customer 190.100.127.128 → 190.100.127.254
Network Id 190.100.64.0/25
Total =128X128 = 16384

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then 32 − 6 = 26. The addresses are

1st customer 190.100.128.1 → 190.100.128.63

Last customer 190.100.159.192 → 190.100.159.254
Network Id 190.100.128.0/26
Total =128X64= 8192

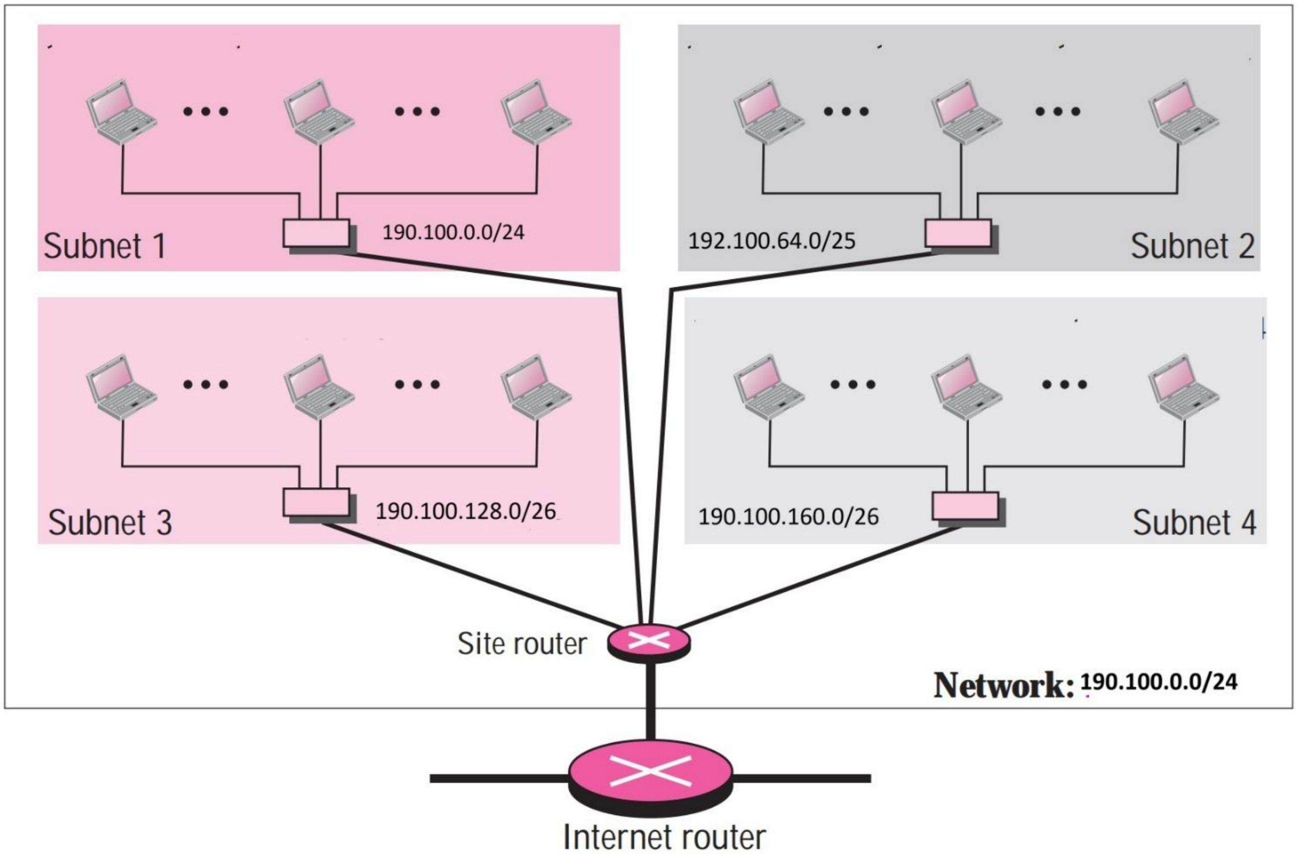
Group 4

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then 32 − 6 = 26. The addresses are

1st customer 190.100.160.1 → 190.100.160.63

Last customer 190.100.223.192 → 190.100.223.254
Network Id 190.100.160.0/26
Total =256X64= 16384

No. of used addresses = 15384 +16384 + 8192 +16384 = 57344.
No. of unused addresses = 65536-57344 = 8192


Subnet 1
190.100.0.0/24
192.100.64.0/25
Subnet 2
Subnet 3
190.100.128.0/26
190.100.160.0/26
Subnet 4
Site router
Network: 190.100.0.0/24
Internet router